

Proof ledger

Repository source: proof/PROOF_LEDGER.md

Status vocabulary

- **ESTABLISHED IN MANUSCRIPT:** a complete derivation is present in the paper, subject to ordinary mathematical review.
- **EXACT-CERTIFIED:** the finite claim is checked with integer and rational arithmetic by the standalone verifier.
- **NUMERICAL AUDIT:** a separate floating-point implementation agrees.
- **RECONSTRUCTION-CHECKED:** an expected derived artifact is rebuilt byte for byte.
- **MUTATION-TESTED:** a targeted corrupted object is rejected.
- **SOURCE-CHECKED:** bibliographic or historical wording has been compared with an identified source.
- **HOSTED-REPLAYED:** the named three-job hosted workflow passed on the stated commit.
- **FRESH-CLONE-REPLAYED:** a new clone of the remote passed the stated replay and ended clean.
- **RELEASE-PENDING:** technically prepared but awaiting a final evidence run, immutable tag, release assets, visibility change, or explicit approval.
- **EXTERNAL REVIEW NEEDED:** no independent subject-matter review has been recorded.

Mathematical claims

ID	Claim	Evidence	Status
M1	Constant-width support identity and diameter-one normalization	Paper Section 2; classical references	ESTABLISHED IN MANUSCRIPT; SOURCE-CHECKED
M2	Blaschke volume-surface identity	Paper Section 2; classical references	ESTABLISHED IN MANUSCRIPT; SOURCE-CHECKED
M3	Minimum circumsphere contact balance and Jung range	Paper Section 2	ESTABLISHED IN MANUSCRIPT
M4	Opposite-point identity and radial support inequality	Paper Lemma 2.1	ESTABLISHED IN MANUSCRIPT
L1	Transport inequality and tangent-gap master inequality	Paper Section 3	ESTABLISHED IN MANUSCRIPT
L2	Rational propagation implications	Paper Section 3; exact lower-cell checks	ESTABLISHED IN MANUSCRIPT; EXACT-CERTIFIED
L3	Four refined cells replace 03A1 with positive margins	Refined NPZ cells and JSON	EXACT-CERTIFIED; NUMERICAL AUDIT
L4	Complete lower partition is gap free and has weakest terminal row 03B02B	Final JSON and exact verifier	EXACT-CERTIFIED; RECONSTRUCTION-CHECKED
N1	Near-Jung four-contact deficiency budget	Paper Section 4	ESTABLISHED IN MANUSCRIPT
N2	Affine contact coordinates and Gram formulas	Paper Section 4	ESTABLISHED IN MANUSCRIPT

ID	Claim	Evidence	Status
N3	Determinant transfer $\det H \geq 1 - S/2 - S^2/2$	Paper Proposition 4.1	ESTABLISHED IN MANUSCRIPT; exact endpoint evaluation
N4	Dominant-edge coordinate-width diagonal bounds	Paper Section 5	ESTABLISHED IN MANUSCRIPT; exact endpoint evaluation
N5	Adjusted 98,306-ray fan is feasible with positive exact margin	Fan NPZ, orbit TSV, exact verifier	EXACT-CERTIFIED; NUMERICAL AUDIT; MUTATION-TESTED
N6	Forced-core volume lower bound	Exact fan volume and determinant transfer	EXACT-CERTIFIED; NUMERICAL AUDIT
N7	Three axis-specific antipodal cap-pair bounds	Hull TSV and cap certificate fields	EXACT-CERTIFIED; NUMERICAL AUDIT; MUTATION-TESTED
N8	Six cap interiors are pairwise disjoint	Paper and exact disjointness fields	ESTABLISHED IN MANUSCRIPT; EXACT-CERTIFIED; MUTATION-TESTED
N9	Near-Jung lower bound exceeds theorem upper enclosure	Final JSON assembly	EXACT-CERTIFIED; NUMERICAL AUDIT
T1	Universal coefficient equals the exact theorem coefficient	split 03B02A/03B02B plus strict near-Jung comparison; exact value in the certificate and paper	EXACT-CERTIFIED; RECONSTRUCTION- CHECKED
T2	Scaling gives the theorem for arbitrary width $d > 0$	Homogeneity	ESTABLISHED IN MANUSCRIPT

Verification and provenance claims

ID	Claim	Evidence	Status
V1	Standalone verifier has no private repository dependency	Path regression test and source inspection	EXACT-CERTIFIED
V2	Final JSON is rebuilt byte identically	Exact verifier and unit test	RECONSTRUCTION- CHECKED
V3	Exact TSV and CSV renderings are rebuilt byte identically	Exact verifier	RECONSTRUCTION- CHECKED
V4	Binary64 implementation agrees on principal quantities	Binary64 transcript	NUMERICAL AUDIT
V5	Base-R implementation agrees on principal quantities	Named-author and hosted R transcripts	NUMERICAL AUDIT
V6	Seven named proof-object mutations are rejected	Mutation transcript	MUTATION-TESTED
V7	Paper-facing constants and lower table match the JSON	Paper reconciliation script	RECONSTRUCTION- CHECKED
V8	Exact-constants appendix matches the JSON	Generator <code>--check</code>	RECONSTRUCTION- CHECKED

ID	Claim	Evidence	Status
V9	Required release documents exist and preserve claim language	Release-document checker	RECONSTRUCTION-CHECKED
V10	Markdown/PDF pairs are complete	Pair checker	RECONSTRUCTION-CHECKED
V11	PDFs are readable, have embedded fonts, and expose text	Hosted Poppler preflight on run 35468019663	HOSTED-REPLAYED
V12	Exact, Python-audit, test, mutation, metadata, and paper paths pass on Ubuntu	Workflow run 35468019663, commit 892d4bc6fad07950e78da66e45b95063a6415af6	HOSTED-REPLAYED
V13	Independent base-R audit passes on hosted R 4.6.1	Workflow run 35468019663	HOSTED-REPLAYED; NUMERICAL AUDIT
V14	Remote fresh clone reproduces all language paths and builds	Named-author transcript for commit 892d4bc6...	FRESH-CLONE-REPLAYED
V15	Fresh-clone working tree remains clean and bundle has complete five-commit history	Fresh transcript and verified transfer bundle	FRESH-CLONE-REPLAYED; RECONSTRUCTION-CHECKED
V16	Hosted/fresh-clone evidence fields and claim boundaries are indexed consistently	certificate/evidence/rc1_remote_audit.json; ci/verify_remote_evidence.py	RECONSTRUCTION-CHECKED

The hosted evidence above refers to <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>.

Publication and review claims

ID	Claim	Status
P1	Focused ten-page Paper v1 manuscript is present	Completed draft
P2	Complete release-document set is present in Markdown and PDF	Completed; 22 paired documents
P3	Final author list and no-affiliation statement are fixed	Completed for v1.1.0
P4	Split license and final citation metadata are fixed	Completed for v1.1.0
P5	Private staging remote exists and the audited candidate was pushed	Completed at commit 892d4bc6...
P6	Hosted Python, R, and document jobs are green on the audited candidate	Completed on run 35468019663
P7	Fresh staging-remote clone replay is recorded	Completed; final status clean
P8	Final v1.1.0 release commit receives its own hosted run	RELEASE-PENDING
P9	Immutable final tag and release assets exist	RELEASE-PENDING
P10	Repository visibility is public and GitHub Release is published	RELEASE-PENDING

ID	Claim	Status
P11	External subject-matter reviewer has checked the theorem	EXTERNAL REVIEW NEEDED
P12	Peer review or journal acceptance exists	Not claimed

Claim boundary

The package does not establish equality, rigidity, sharpness, a minimizing body, or Meissner extremality. It should not be described as independently verified or formally verified while P11 remains open. The named-author fresh clone and hosted CI are reproducibility evidence, not external mathematical review.

Current bottleneck

The mathematical, finite-certificate, authorship, citation, licensing, hosted-pre-release, fresh-clone, README-rendering, and Table 3 presentation gates are complete. The remaining publication bottlenecks are:

1. green hosted jobs on the final `v1.1.0` release commit;
2. final named-author approval of the complete paper, release notes, and announcement wording;
3. an annotated `v1.1.0` tag and green tag-triggered workflow;
4. a clean replay from a fresh checkout of the tag;
5. deterministic release assets and independent post-download hash verification;
6. explicit approval of the visibility change and GitHub Release; and
7. external mathematical review after release.